# CONVERGENCE OF SEM

- FOR STATIONARY PB (HELMHOLTZ)

$$\| u - u_h \|_{L^2(0,L)} \lesssim h^{p+1} \qquad p = \text{polynomial degree}$$

$$\| u - u_h \|_{H^1(0,L)} \lesssim h^{p}$$

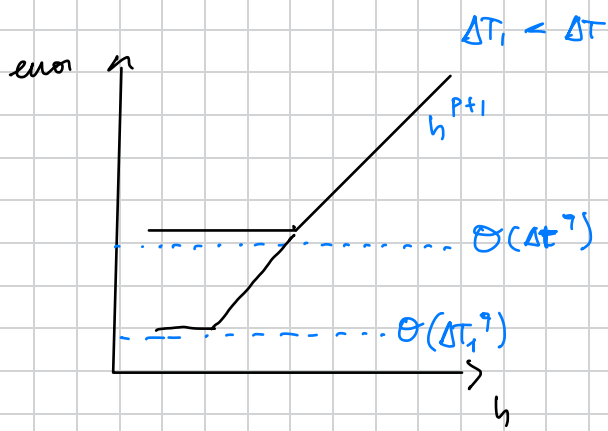
- FOR NON-STATIONARY PB (WAVE EQ)

$$\| u - u_h \|_{L^2(0,L)}(T) \lesssim h^{p+1} + \Delta t^{q}$$

$$\| u - u_h \|_{H^1(0,L)}(T) \lesssim h^{p} + \Delta t^{q}$$

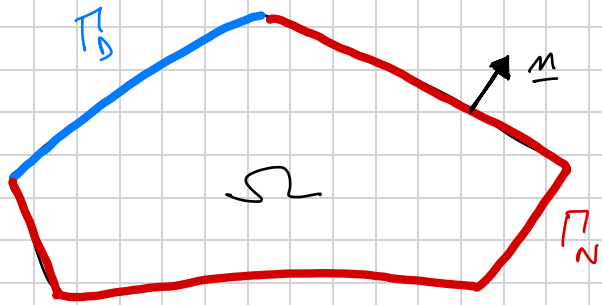
SPACE DISCR. TIME INTEGRATION

LEAP-FROG $q = 2$

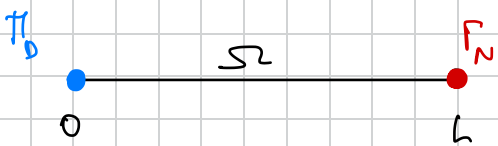


# SEM FOR 2D/3D PB.

$$
\begin{cases}
\rho \dfrac{\partial^2 u}{\partial t^2} - \underbrace{\nabla \cdot}_{\text{DIVERGENCE}} \left( \mu \underbrace{\nabla u}_{\text{GRAD}} \right) = f & \text{in } \Omega \times (0,T] \\
u(\underline{x},0) = u_0(\underline{x}) & \text{in } \Omega \\
\dfrac{\partial u}{\partial t}(\underline{x},0) = \underline{v}_0(\underline{x}) & \text{in } \Omega \\
u(\underline{x},t) = 0 & \text{on } \Gamma_D \times (0,T] \\
\mu \nabla u \cdot \underline{n} = g_N(\underline{x},t) & \text{on } \Gamma_N \times (0,T]
\end{cases}
$$




$\Gamma_D \cap \Gamma_N = \emptyset$ $\qquad$ $\Gamma_D \cup \Gamma_N = \partial\Omega$ (BOUNDARY)




WEAK FORM

$$\int_\Omega \rho \frac{\partial^2 u}{\partial t^2} v \, dx - \underbrace{\int_\Omega \nabla \cdot (\mu \nabla u) v \, dx}_{\text{INTEGRATION BY PARTS}} = \int_\Omega f v \, dx$$

DIV - THM $$\int_\Omega \nabla \cdot \underline{a} = \int_{\partial\Omega} \underline{a} \cdot \underline{n}$$

$$\underline{a} = \underline{b} \varphi \rightarrow \underbrace{\int_\Omega \nabla \cdot (\underline{b} \varphi)} = \int_{\partial\Omega} \underline{b} \cdot \underline{n} \varphi$$

$$\int_\Omega \nabla \cdot (\underline{b} \varphi) = \underbrace{\int_\Omega \nabla \cdot \underline{b} \varphi}_{\int_\Omega \nabla \cdot (\overset{\underline{b}}{\mu \nabla u}) \overset{\varphi}{v}} + \int_\Omega \underline{b} \cdot \nabla \varphi$$

$$- \int_\Omega \nabla \cdot (\mu \nabla u) v = \int_\Omega \mu \nabla u \cdot \nabla v - \int_{\partial\Omega} \mu \nabla u \cdot \underline{n} v$$

$$\int_\Omega \rho \frac{\partial^2 u}{\partial t^2} v \, dx + \int_\Omega \mu \nabla u \cdot \nabla v \, dx$$

$$\underbrace{- \int_{\partial\Omega} \mu \nabla u \cdot n v \, dx}_{} = \int_\Omega f v \, dx$$

$$\underbrace{- \int_{\Gamma_D} \mu \nabla u \cdot \cancelto{0}{n} v \, dx}_{v = 0 \text{ on } \Gamma_D} - \int_{\Gamma_N} \underbrace{\mu \nabla u \cdot \underline{n}}_{g_N} v \, dx$$

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$$\underbrace{\int_\Omega \rho \frac{\partial^2 u}{\partial t^2} v \, dx}_{m\left(\frac{\partial^2 u}{\partial t^2}, v\right)} + \underbrace{\int_\Omega \mu \nabla u \cdot \nabla v}_{a(u,v)} = \underbrace{\int_\Omega f v \, dx + \int_{\Gamma_N} g_N v \, dx}_{F(v)} \qquad (1)$$

$$H^1_{\Gamma_D}(\Omega) = \left\{ v : \Omega \to \mathbb{R} \text{ s.t. } v \in H^1(\Omega), \ v = 0 \text{ on } \Gamma_D \right\}$$

$$\| v \|^2_{H^1(\Omega)} = \| v \|^2_{L^2(\Omega)} + \left\| \frac{\partial v}{\partial x} \right\|^2_{L^2(\Omega)} + \left\| \frac{\partial v}{\partial y} \right\|^2_{L^2(\Omega)} < +\infty$$

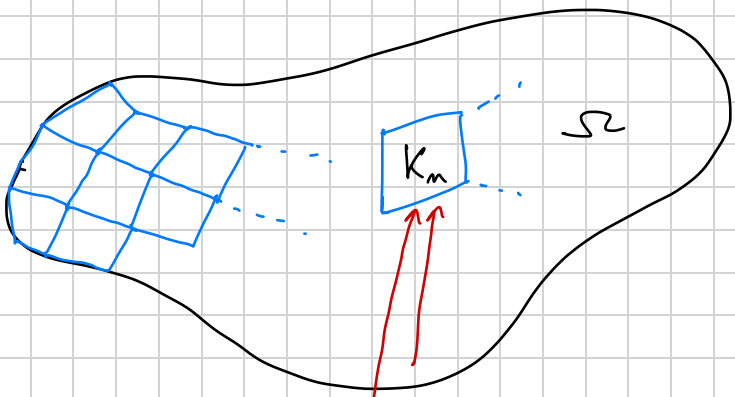
WEAK FORM :

$$\begin{cases} \forall t \in (0,T] \text{ find } u \in H^1_{\Gamma_D}(\Omega) \text{ s.t. } (1) \\ \text{holds } \forall v \in H^1_{\Gamma_D}(\Omega). \\ + \text{ INITIAL CONDITIONS} \end{cases}$$

SE FORMULATION

Fix $V_h \subset V = H^1_{\Gamma_D}(\Omega)$ $\quad \dim(V_h) < +\infty$

We need to properly define $V_h$

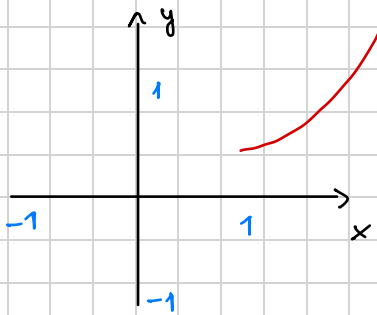



$\mathcal{T}_h$ is a mesh for $\Omega$ made by

- quads if $\Omega \subseteq \mathbb{R}^2$
- hexahedrals $\Omega \subseteq \mathbb{R}^3$

$$\mathcal{T}_h = \bigcup_{m=1}^{N_h} K_m$$

REF. EL IN 2D

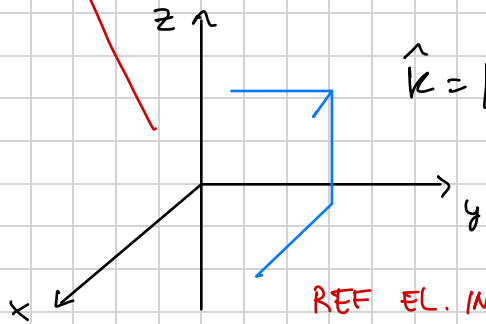



$$\hat{K} = [-1,1]^2$$

$$K_m = F_m(\hat{K})$$

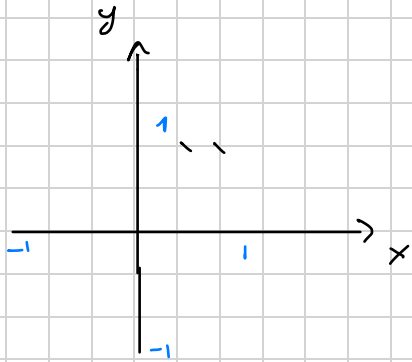
↳ BILINEAR MAP (2D)

↳ TRILINEAR MAP (3D)




$$\hat{K} = [-1,1]^3$$

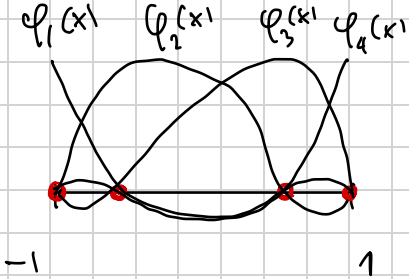
REF EL. IN 3D

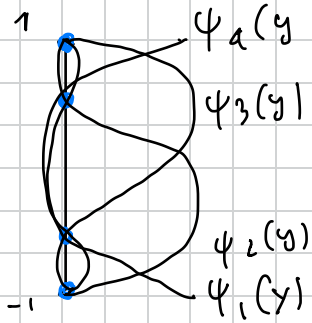



● GLL NODES X-DIR

● GLL NODES Y-DIR

I define basis functions in $\hat{K}$ based on tensor product GLL nodes



$n+1$ points $x$

$n+1$ points $y$

$\Big\}$ $(n+1)^2$ points in $\hat{K}$

$(n+1)^2$ basis functions

$$\{\phi_k\}_{k=1}^{(n+1)^2} = \{\phi_k(x,y) = \varphi_i(x)\,\psi_j(y)\}_{i,j=1}^{n+1}$$

$$k \longleftrightarrow (i,j)$$

$$V_h = \left\{ v \in C^0(\Omega) : v\big|_{K_m} \in \mathbb{Q}^n(K_m) \ \forall K_m \in \mathcal{T}_h \ \ v \equiv 0 \text{ on } \Gamma_D \right\}$$

$\mathbb{Q}^n$ is the space of polynomials of degree $\leq n$ in each spatial dimension

Ex: $n=3$

x - dir $\{1, x, x^2, x^3\}$

y - dir $\{1, y, y^2, y^3\}$

$$\mathbb{Q}^3 = \text{span}\{\overset{1}{1}, \overset{2}{x}, \overset{3}{x^2}, \overset{4}{x^3}, \overset{5}{y}, \overset{6}{y^2}, \overset{7}{y^3}, \overset{8}{xy}, \overset{9}{xy^2}$$

$$\underset{10}{xy^3}, \underset{11}{x^2y}, \underset{12}{x^2y^2}, \underset{13}{x^2y^3}, \underset{14}{x^3y}, \underset{15}{x^3y^2}$$

$$x^3 y^3 \}$$

T6

$$u_h(\underline{x},t) = \sum_{j=1}^{M} u_j(t)\,\phi_j(\underline{x})$$

$\underline{x} = (x,y)$

$\underline{x} = (x,y,z)$

SUBSTITUTE INTO THE WEAK FORM

$\forall t \in [0,T]$ find $u_h(t) \in V_h$ s.t.

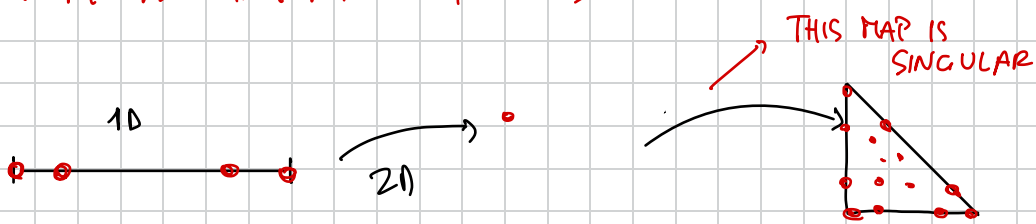
$$m(\ddot{u}_h, v_h) + a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h$$

$+$ I.C.

$$\underline{U}(t) = (u_h^1(t), u_h^2(t) \ldots \ldots u_h^M(t))^T$$

$$M\,\ddot{\underline{U}}(t) + A\,\underline{U}(t) = \underline{F}(t)$$

M IS DIAGONAL BY CONSTRUCTION




DUFFY TRANSFORMATION

OTHER BOUNDARY CONDITIONS

1) $\nabla u \cdot \underline{n} + \frac{1}{z} \frac{\partial u}{\partial t} = 0$ $\quad z = c$ ABSORBING CONDITIONS

$z > 0$

2) PML CONDITIONS

3) PERIODIC CONDITIONS

4) ROBIN CONDITIONS

$$\begin{cases} M \ddot{\underline{U}} + A \underline{U} = \underline{F}(t) \\ \underline{U}(0) = \underline{U}_0 \\ \dot{\underline{U}}(0) = \underline{V}_0 \end{cases}$$

$\rightarrow$ LEAP-FROG

NEWMARK SCHEME

$$\ddot{\underline{U}} = M^{-1} ( \underline{F} - A \underline{U} ) = \mathcal{L}(\underline{U}, t)$$

$$\dot{\underline{U}} = \underline{Z}$$

$$\begin{cases} \dot{\underline{U}} = \underline{Z} \\ \dot{\underline{Z}} = \mathcal{L}(\underline{U}, t) \end{cases}$$

NOTATION

$$\underline{U}(t_k) = \underline{U}_k$$

$$\mathcal{L}_k = \mathcal{L}(U_k, t_k)$$

NEWMARK UPDATE

$$\begin{cases} U_{k+1} = U_k + \Delta t\, Z_k + \Delta t^2 \left( \beta \mathcal{L}_{k+1} + \left(\frac{1}{2} - \beta\right) \mathcal{L}_k \right) \\ Z_{k+1} = Z_k + \Delta t \left( \gamma \mathcal{L}_{k+1} + (1-\gamma) \mathcal{L}_k \right. \end{cases}$$

$$0 \leq \gamma \leq 1 \qquad 0 \leq 2\beta \leq 1$$

MATLAB

- $\gamma = \frac{1}{2}$ $\quad \beta = \frac{1}{4}$ $\Rightarrow$ IMPLICIT (NO CFL) 2ND ORDER SCHEME $(\mathcal{O}(\Delta t^2))$
- $\gamma \neq \frac{1}{2}$ $\Rightarrow$ FIRST ORDER $(\mathcal{O}(\Delta t))$
- $\beta = 0$ $\Rightarrow$ EXPLICIT SCHEME

$$\rho u_{tt} - (\mu u_x)_x = f \quad \leftarrow \text{2ND ORDER WAVE EQ}$$

$$\begin{cases} w_1 = \rho u_t \\ w_2 = \mu u_x \end{cases} \qquad (w_1)_t - (w_2)_x = f$$

$\rho, \mu$ constant

$$(w_2)_t = (\mu u_x)_t = \mu u_{xt} = \mu u_{tx} = \frac{\mu}{\rho}(w_1)_x$$

$$(w_1)_x = (\rho u_t)_x = \rho u_{tx}$$

$$\begin{cases} (w_1)_t - (w_2)_x = f \\ (w_2)_t - \frac{\mu}{\rho}(w_1)_x = 0 \end{cases} \leftarrow \text{SYSTEM OF FIRST WAVE EQ.}$$

$$\underline{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \rightarrow \underline{w}_t - A\underline{w}_x = \underline{f} \leftarrow (*)$$

$$\underline{f} = \begin{pmatrix} f \\ 0 \end{pmatrix} \qquad A = \begin{pmatrix} 0 & -1 \\ -\frac{\mu}{\rho} & 0 \end{pmatrix}$$

HYPERBOLIC PB.

$$\text{eig}(A) \Rightarrow \det(A - \lambda I) = 0$$

$$\lambda^2 - \frac{\mu}{\rho} = 0 \qquad \lambda_{12} = \pm c$$

REAL & DISTINCT EIGENVALUES

(*) + INITIAL CONDITIONS

+ BOUNDARY CONDITIONS

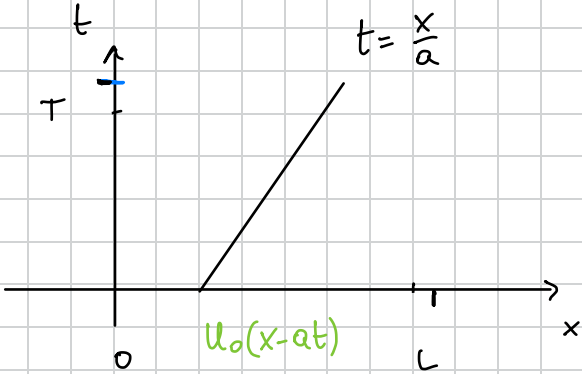
$\Big\}$ FOR $\underline{u_t}$ & $\underline{u_x}$

$\underline{w_1}$ & $\underline{w_2}$

SIMPLEST HYP PB. $\lambda = a > 0$

$$\begin{cases} u_t + a\, u_x = 0 & x \in [0,L] \quad t \in (0,T] \\ u(x,0) = u_0(x) \end{cases}$$

TRANSPORT EQUATION



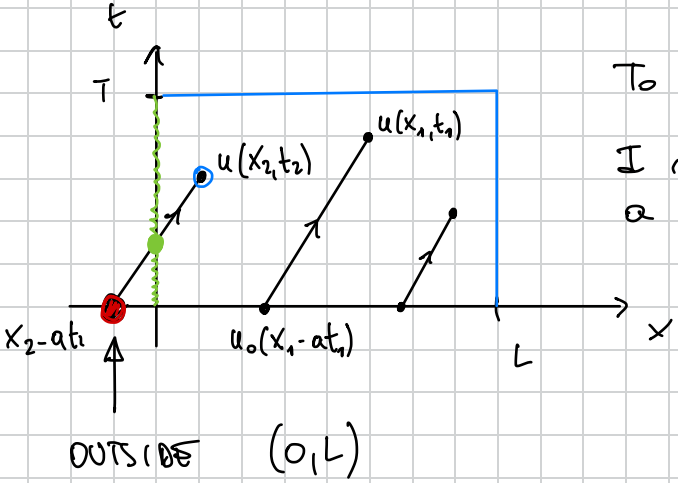

$$u(x,t) = u_0(x - at)$$

CHARACTERISTIC CURVE

$X(t)$ : $x_0 + at$ solution to $\begin{cases} \dfrac{dX}{dt} = a \\ X(0) = x_0 \end{cases}$

$$\frac{d}{dt}\, \underbrace{u(X(t),t)} = \frac{du}{dX}\frac{dX}{dt} + \frac{du}{dt} = u_x\, a + u_t = 0$$

$u(X(t),t)$ is constant along $X(t)$




To compute $u(x_2,t_2)$

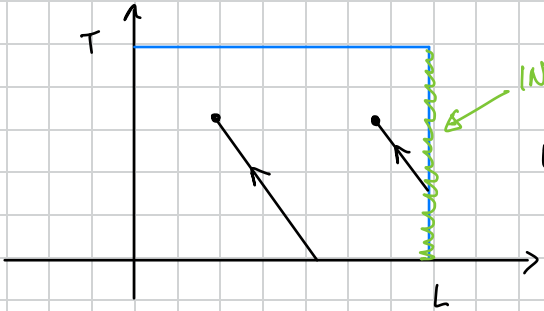
I need to introduce a boundary condition on $\underbrace{\{0\}\times(0,T]}_{\text{INFLOW BOUNDARY } (a>0)}$

$$\Downarrow$$

$$u(0,t) = \varphi(t)$$

$$\begin{cases} u_t + a u_x = 0 & x\in(0,L) \quad t\in(0,T] \\ u(0,t) = \varphi(t) & t\in(0,T] \\ u(x,0) = u_0(x) & x\in(0,L) \end{cases} \qquad (P)$$

$a<0$




INFLOW BOUNDARY $\{L\}\times(0,T]$

$$u(L,t) = \psi(t)$$